

IANMANTECH SOLUTIONS

Problem Statement : Project VIPER

Enterprise On-Premises Voice Intelligence Platform

1. Context & Background

Modern enterprises in regulated industries (finance, healthcare, legal, defense) lose vast amounts of critical business intelligence shared during daily voice conversations, client meetings, and depositions. While cloud-based AI meeting assistants (such as Otter, Plaud, or Granola) exist, they require routing sensitive audio, transcripts, and employee metrics through public cloud infrastructure. For security-conscious organizations, this creates unacceptable compliance risks, corporate espionage vulnerabilities, and data residency violations.

Project VIPER bridges this gap. VIPER is an enterprise-grade, fully on-premises / air-gapped hardware and software ecosystem designed to capture, process, index, and automate voice intelligence completely behind customer firewalls. Utilizing dedicated wearable recording hardware (the H-R191 card/pendant line), the platform processes end-to-end voice pipelines—from Bluetooth Low Energy (BLE) ingestion to local Speech-to-Text (STT), automated PII redaction, corporate knowledge vector indexing, and CRM/Email action agents—without a single byte leaving the corporate network.

2. Problem Statement

Build a secure, privacy-first, on-premises Voice Intelligence Platform (VIPER) that ingests meeting audio from dedicated hardware/companion interfaces, enforces automated compliance and PII redaction, indexes transcripts into an ACL-aware enterprise knowledge graph, and automatically executes post-meeting workflows via local agents—all operating without internet connectivity.

3. Key Pillars & Product Scope

Participants must prioritize functionality based on enterprise willingness-to-pay and compliance necessity:

Pillar 1: Automated Compliance & Security Guardrails (MVP Core)

- **Local Ingestion & Redaction:** Process incoming audio (WAV/OPUS) and run local STT + Speaker Diarization. Automatically detect and redact PII, PHI, and PCI data (names, national IDs, financial accounts) prior to making transcripts searchable.

- **Policy Breach Classifier:** Identify policy violations (e.g., missing consent preambles, restricted location recording).
- **Auditability & Controls:** Maintain an immutable, append-only audit log tracking every audio access, export, and query.

Pillar 2: Corporate Knowledge Brain & RAG Engine

- **ACL-Aware Vector Indexing:** Automatically chunk redacted transcripts and index them into an on-premises vector database.
- **Role-Based RAG Retrieval:** Enable natural language querying over past meetings while strictly enforcing user access control lists (ACLs) (e.g., legal discussions are restricted from unauthorized roles).
- **Source Citation:** RAG responses must provide traceable citations linked to exact audio timestamps, speaker IDs, and device serial numbers.

Pillar 3: Post-Meeting Agent Management

- **Action Item Extraction:** Parse transcripts to identify action items and match assigned owners against an internal corporate directory.
- **Human-in-the-Loop Automation:** Draft follow-up emails and CRM tasks (e.g., Salesforce/Exchange-style JSON payloads) placed into a user approval queue before execution.

4. Hardware & Companion Integration (H-R191 Ecosystem)

Your solution must emulate or interface with the **H-R191 AI Smart Recorder** hardware line:

- **Hardware Profile:** Credit-card form factor (86 \times 53 \times 3.2\text{ mm}), 30\text{ g} mass, dual MEMS microphones, 400\text{ mAh} battery, magnetic charging).
- **Connectivity & Bandwidth Constraints:** Operates over BLE 5.4. Bulk audio transfer is rate-limited ($\sim 50\text{ KB/s}$ for OPUS offload).
- **Companion App Integration:** A mobile companion agent (iOS/Android facade via viper-edge-sdk) handles BLE pairing, clock synchronization with on-prem servers, offline chunk buffering, and checksum verification before clearing on-device flash.

5. Technical Requirements & System Constraints

Category	Requirement Specification
Deployment Model	Fully Air-Gapped / Private Cloud. Zero calls to public SaaS/APIs (OpenAI, Anthropic, cloud STT).

AI/ML Stack	Local open-weight models only (e.g., Whisper/faster-whisper for STT; Llama/Qwen/Mistral for LLM; local Sentence-Transformers for embeddings).
Datastores	Local Object Store (S3-compatible/MinIO), Relational DB (PostgreSQL) for ACL/metadata, and Vector DB (pgvector/Qdrant/Milvus).
Security	OIDC/SAML authentication support, customer-managed encryption at rest (AES-256), and mTLS inter-service communications.
Performance Target	Ingest-to-redacted-transcript processing latency $\leq 0.3 \times$ real-time on local GPU/CPU execution.

6. Evaluation & Judging Criteria

1. **Air-Gap Integrity & Privacy (25%):** System functions completely offline. Sensitive audio/PII is fully redacted before reaching vector indexes or external interfaces.
2. **Accuracy & AI Quality (25%):** Quality of STT transcription, diarization, PII entity recognition, accurate RAG retrieval, and strict citation back-linking.
3. **Workflow & Agent Automation (20%):** Robustness of post-meeting action item extraction and human-in-the-loop task generation.
4. **Hardware/SDK Integration Experience (15%):** Simulation or handling of BLE file offload constraints, sync protocols, checksum validation, and device state management.
5. **Architectural Elegance & Security (15%):** Clean microservices/layered design, strict role-based access control (RBAC/ACL), and audit log implementation.

7. Expected Deliverables

- **Functional System Prototype:** Running web console + backend pipeline demonstrating full processing flow from raw file ingestion to redacted player view.
- **RAG & Agent Console:** Natural language search interface enforcing ACLs with timestamped audio citations, along with an action item approval queue.
- **Device Gateway Simulator:** Script/app simulating the H-R191 BLE/companion sync flow (transferring OPUS files, verifying checksums, updating device status).
- **Architecture Diagram & Demo Video:** Documenting internal data boundaries, local model registry setup, and zero-internet compliance verification.

Thanks & Regards